

ZecurX LABS: Internal Product Brief

1.0 Platform Vision and Core Purpose

ZecurX LABS is a gamified cybersecurity platform designed to function as both a competitive environment and an educational tool. At its core, it is a Capture The Flag (CTF) competition, where participants engage with a variety of security-related challenges to find hidden "flags"—unique text strings that prove a challenge has been successfully solved. This approach transforms complex cybersecurity concepts into an interactive, treasure-hunt-like experience.

The strategic importance of this platform lies in its ability to engage, train, and assess individuals in practical, hands-on cybersecurity skills. By moving beyond theoretical knowledge, ZecurX LABS provides a dynamic environment for users to apply and refine their abilities in areas like web exploitation, cryptography, and forensics. This brief will provide a detailed examination of the platform's features and architecture, beginning with the core user journey that drives engagement.

2.0 The Core User Experience: CTF Gamification Loop

The core user experience is built around a central engagement loop designed to be intuitive and rewarding. A clear, gamified progression is vital for user retention and sustained skill development, encouraging participants to move from browsing challenges to solving them and ultimately competing for a top spot on the leaderboard. This loop forms the fundamental workflow for every user on the platform:

1. **Browse Challenges:** The user selects a challenge from a comprehensive list.
2. **Read the Description:** They review the problem statement to understand the objective.
3. **Solve the Puzzle:** The user applies their technical skills to solve the challenge.
4. **Find the Flag:** They locate the hidden flag, typically in the format CTF{something_here}.
5. **Submit the Flag:** The flag is entered into the platform to earn points.
6. **Compete on Leaderboard:** The user's score is updated, and their rank is adjusted accordingly.

2.1 Main Dashboard: The User's Command Center

Upon logging in, the user is greeted by the Main Dashboard (/dashboard), which serves as a command center providing an immediate and actionable overview of their progress and standing within the platform.

Component	Function
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Welcome Header	Provides a personalized greeting to the user.
Notification Bell	Alerts the user to unread announcements or messages with a count indicator.
Statistics Cards	Displays key metrics at a glance: total score, current rank, and total challenges solved.
Category Progress	Uses visual bars to show the user's completion progress within each challenge category.
Recent Challenges	Offers quick access to available challenges, encouraging immediate engagement.
Leaderboard Preview	Showcases the top five players, fostering a sense of competition.
Quick Links	Allows for easy navigation to primary sections like the full challenges page and user profile.

2.2 Challenge Engagement

Users interact with challenges through two primary interfaces: the Challenges Page for discovery and the Challenge Detail Page for solving.

Challenge Discovery

The Challenges Page (/challenges) acts as the central hub for finding new tasks. It is equipped with several features to help users navigate the content effectively:

- **Category Filters:** Users can filter the challenge list by type (e.g., Web, Crypto) to focus on specific areas of interest.
- **Challenge Cards:** Each challenge is presented on a card displaying its title, point value, and difficulty.
- **Difficulty Badges:** Color-coded badges—green for Easy (50-75 points), yellow for Medium (100-125 points), and red for Hard (150+ points)—allow users to quickly assess a challenge's complexity and potential reward.
- **Solved Indicators:** A checkmark appears on challenges the user has already completed, providing a clear visual record of their progress.

Challenge Solving

The Challenge Detail Page (/challenges/:id) is where the core problem-solving takes place. It features a full description of the problem, a field for flag submission, and a hint system. This system adds a strategic layer to the experience, as users can unlock hints at the cost of 5 to 40 points, depending on the hint. The point deduction is applied

immediately upon unlocking a hint, forcing users to weigh the benefit of assistance against its impact on their final score.

The platform offers a diverse set of challenges across several key cybersecurity domains:

Category	Description
Web Exploitation	Challenges involving website vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), and authentication bypass.
Cryptography	Tasks focused on codes and ciphers, ranging from classical ciphers to modern encryption and hash cracking.
Forensics	Investigative challenges that involve file analysis, memory dumps, and finding hidden data in logs or metadata.
Binary Exploitation	Challenges related to compiled programs, including buffer overflows, format string vulnerabilities, and reverse engineering.
General Skills	Puzzles that test foundational skills required for CTFs, such as Linux commands, basic scripting, and research abilities.

2.3 Competitive and Progression Elements

To drive long-term engagement, ZecurX LABS incorporates robust competitive and progression mechanics.

The Leaderboard The Leaderboard (/leaderboard) is the primary competitive hub, displaying a ranked list of all users based on their total score. It includes the user's **Rank**, **Username**, **Score**, and the total number of **Challenges Solved**. To enhance the competitive dynamic, the top three players are highlighted with special styling, and the current user's own rank is always highlighted for easy reference.

User Profile & Achievements The User Profile page (/profile) provides a detailed summary of an individual's statistics and accomplishments. It includes an overview of their score and rank, along with an achievements system that awards badges for reaching specific milestones. This system provides clear goals for users to strive for beyond simply solving the next challenge. Key achievements include:

- **First Blood:** Awarded for being the first user on the platform to solve any given challenge.
- **Century:** Unlocked upon reaching a total score of 100 points.
- **Half K:** Unlocked upon reaching a total score of 500 points.
- **Completionist:** Awarded for solving all available challenges on the platform.

While the main CTF provides a compelling environment for self-directed learning, the platform also offers a more structured, curriculum-based experience through the Student Portal.

3.0 Student Portal: Structured Learning Environment

The Student Portal is a distinct learning management system integrated within ZecurX LABS, designed to provide a more guided, curriculum-based educational experience. In contrast to the open CTF environment, the Student Portal allows administrators to organize challenges into structured courses and modules, offering a clear learning path for students. This structure makes the portal an ideal solution for academic courses, corporate training programs, or guided onboarding for new cybersecurity professionals.

Access to this environment is managed through a simple enrollment process. An administrator first creates a course and then generates a unique enrollment code for a student. The student uses this code to join the course and gain access to its specific content.

3.1 Student-Facing Features

The Student Portal is organized into several key areas, each tailored to the student learning journey.

1. Student Dashboard (/student/dashboard)

- This page serves as a personalized hub, displaying an overview of enrolled courses, overall progress, and earned achievements. It also features a recent activity feed to keep students updated on their learning path.

2. Course Navigation (/student/courses, /student/course/:id)

- The "My Courses" page presents a grid of all courses a student is enrolled in, each with a progress bar and a custom color set by the administrator. From here, students can navigate into a specific course to view its modules.

3. Module and Challenge Interaction

(/student/course/:courseid/module/:moduleid, /student/challenge/:id)

- Within a course, content is broken down into modules, each containing a list of challenges. Student-specific challenges differ from standard CTF challenges by including features such as contextual descriptions relevant to the course material and support for **multiple flag submissions**, allowing for more complex, multi-part problems.

This structured environment provides a powerful tool for guided education, which is managed and controlled through a comprehensive set of administrative tools.

4.0 Administrative Capabilities: Platform Management and Control

The Admin Panel is the centralized backend for managing all aspects of the ZecurX LABS platform. Access to all administrative endpoints is restricted to users with an admin role, ensuring secure control over platform content, users, and settings. The panel's capabilities are organized into distinct management areas.

Management Area	Primary Route	Key Capabilities
User Management	/admin/users	View all users; create, edit, or delete user accounts; reset a user's progress; and promote a user to an admin role.
Challenge Management	/admin/challenges	Create, edit, and delete standard CTF challenges; set point values, difficulty, and categories; and add multiple hints with associated point costs.
Category Management	/admin/categories	Create, edit, and delete challenge categories, including setting names, descriptions, and icons.
Student Portal Management	/admin/student-portal	Manage all aspects of the structured learning environment, including creating courses, modules, and student-specific challenges with multi-flag support. Admins can also generate enrollment codes and manage student enrollments.
Submissions View	/admin/submissions	View a log of all flag submissions across the platform, with the ability to filter by correct or incorrect attempts and see which users attempted specific challenges.
Notifications	/admin/notifications	Send platform-wide announcements or direct messages to specific users, and view a history of all previously sent notifications.

4.1 Notification System

The platform includes a dual-function notification system to facilitate communication between administrators and users.

User-Facing Features:

- A **bell icon** in the main header displays a count of unread notifications.

- Users can mark Clicking the icon reveals a dropdown list of recent notifications, showing the message and a relative timestamp (e.g., "5m ago").
- individual notifications as read.
- A "Mark all as read" function is available to clear all notifications at once.

Admin-Facing Features:

- Admins can compose and send **announcements** that are broadcast to all users on the platform.
- Admins have the ability to send **direct messages** to specific users, using a search function to find recipients by username or email.
- A complete history of sent notifications is available for review.

These robust features are supported by a modern and secure technical architecture.

5.0 Technical Architecture and Implementation

ZecurX LABS is a modern web application built with a decoupled frontend and backend, and the platform is currently fully functional. This architecture separates the user interface from the server-side logic, allowing for independent development and scaling.

Layer	Technology	Purpose
Frontend	React 18 + Vite	A modern framework and build tool for creating a fast, interactive user interface.
	Tailwind CSS / Shadcn/UI	Provides a utility-first CSS framework and a component library for building a consistent and responsive design.
	React Router	Manages client-side navigation and routing within the single-page application.
	Sonner (Toast Notifications), Lucide React (Icons)	Provides immediate user feedback and a consistent visual language.
Backend	Python + FastAPI	A high-performance web framework for building APIs, chosen for its speed and asynchronous capabilities.
	Motor	An asynchronous Python driver for MongoDB, enabling non-blocking database operations.
	PyJWT / bcrypt	Libraries used for handling JSON Web Token (JWT) authentication and securely hashing

		passwords.
Database	MongoDB	A NoSQL document database that offers flexibility and scalability for storing platform data.

5.1 Authentication and Security

The platform incorporates several industry-standard security measures to protect user data and control access.

- **Password Security:** All user passwords are computationally secured using the bcrypt hashing algorithm with a salt, preventing them from being stored in plaintext.
- **Session Management:** User sessions are managed via JSON Web Tokens (JWT) with a 7-day expiry. The token is transmitted in a Bearer token header with every authenticated API request.
- **Access Control:** The backend enforces role-based access control using middleware, which restricts sensitive administrative routes to authorized users only.
- **Input Validation:** All incoming data to the backend is validated using Pydantic models to protect against common vulnerabilities like injection attacks and ensure data integrity.

5.2 Database Schema

The platform utilizes MongoDB as its document database, which organizes data into collections rather than traditional tables. The key collections that form the platform's data structure include:

- users
- categories
- challenges
- user_progress
- courses
- course_modules
- student_challenges
- student_enrollments
- enrollment_codes
- notifications

This schema supports all current features and is designed to accommodate future development plans.

6.0 Development Roadmap

The future direction of the ZecurX LABS platform is focused on enhancing the user experience, expanding interactive capabilities, and introducing new competitive formats. Development initiatives are separated into immediate priorities for the next update cycle and longer-term strategic goals.

Planned for Next Update

Feature	Description
Docker Labs	Provides isolated, containerized environments for users to interact with more complex challenges safely.
OAuth2 Login	Allows users to register and sign in using third-party accounts such as Google or GitHub.
Profile Editing	Enables users to update their own profile information directly through the user interface.
Password Reset	Implements an email-based password recovery system for users who have forgotten their credentials.

Long-term Roadmap

- **Team Competitions:** Introduce functionality for users to form teams and compete collaboratively in CTF events.
- **Write-up Submissions:** Allow users to submit and share their solutions and methodologies after solving a challenge.
- **Time-limited CTFs:** Host scheduled, time-bound competition events with unique challenge sets.
- **Premium Subscription Tier:** Develop a premium tier offering access to exclusive content, advanced features, or private competitions.
- **Challenge File Downloads:** Enable challenges that require users to download and analyze files locally.
- **Real-time Scoreboards:** Implement live updates to the leaderboard for a more dynamic competition experience.